Q1.

A group of friends want to prepare a meal. They start preparing the meal at 6:30pm

Activities to prepare the meal are shown in **Figure 1** below.

Figure 1

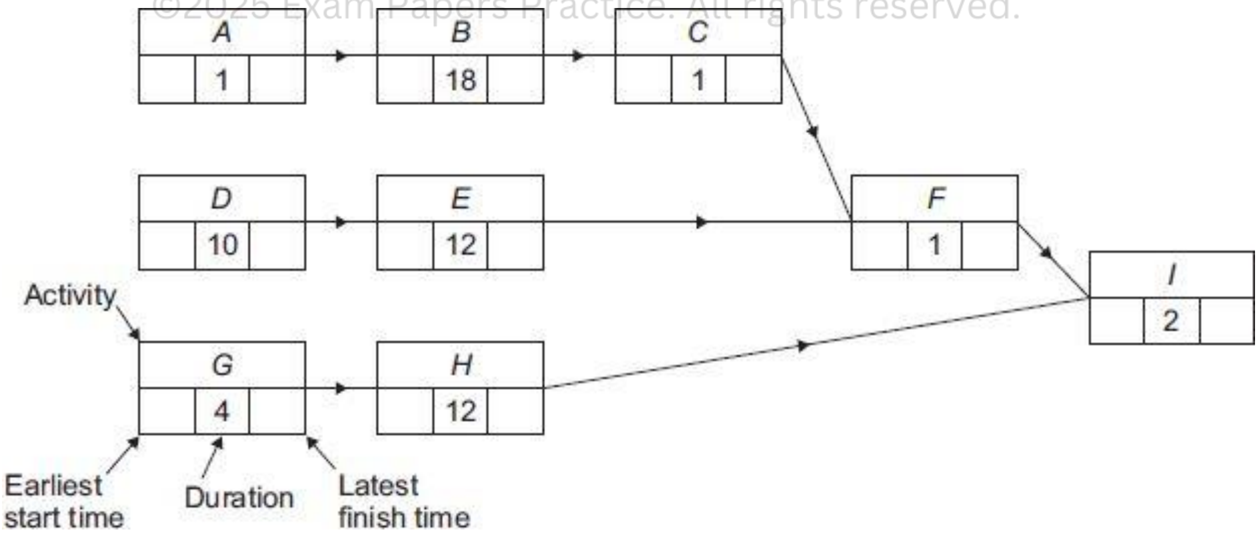
Label	Activity	Duration (mins)	Immediate predecessors
A	Weigh rice	1	–
B	Cook rice	18	*A*
C	Drain rice	1	*B*
D	Chop vegetables	10	–
E	Fry vegetables	12	
F	Combine fried vegetables and drained rice	1	
G	Prepare sauce ingredients	4	–
H	Boil sauce	12	
I	Serve meal on plates	2	

(a) (i) Use **Figure 2** shown below to complete **Figure 1** above.

(1)

(ii) Complete **Figure 2** showing the earliest start time and latest finishing time for each activity.

Figure 2



(2)

(b) (i) State the activity which must be started first so that the meal is served in the shortest possible time.

Fully justify your answer.

(2)

(ii) Determine the earliest possible time at which the preparation of the meal can be completed.

(1)

(c) The group of friends want to cook spring rolls so that they are served at the same time as the rest of the meal. This requires the additional activities shown in **Figure 3**.

Figure 3

Label	**Activity**	**Duration**	**Immediate predecessors**
J	Switch on and heat oven		–
K	Put spring rolls in oven and cook		
L	Transfer spring rolls to serving dish		

It takes 15 seconds to switch on the oven.

The oven must be allowed to heat up for 10 minutes before the spring rolls are put in the oven.

It takes 15 seconds to put the spring rolls in the oven.

The spring rolls must cook in the hot oven for 8 minutes.

It takes 30 seconds to transfer the spring rolls to a serving dish.

(i) Complete **Figure 3** above.

(1)

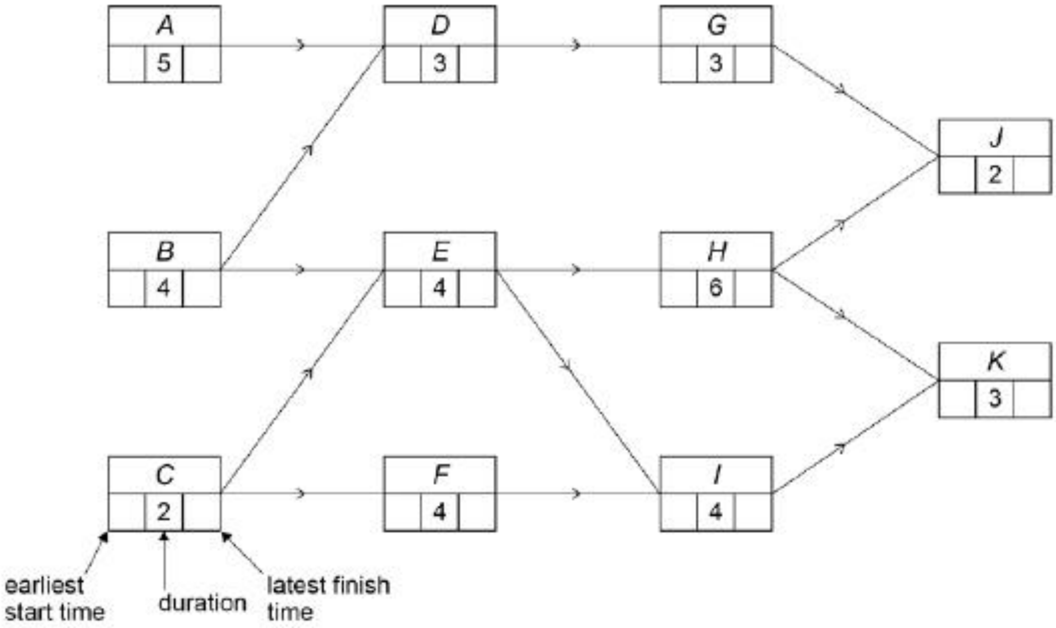
(ii) Determine the latest time at which the oven can be switched on in order for the spring rolls to be served at the same time as the rest of the meal.

(2)

(Total 9 marks)

Q2.

Deva Construction Ltd undertakes a small building project. The activity network for this project is shown below, where each activity's duration is given in hours.



(a) Complete the activity network for the building project.

(2)

(b) Deva Construction Ltd is able to reduce the duration of a single activity to 1 hour by using specialist equipment.

State, with a reason, which activity should have its duration reduced to 1 hour in order to minimise the completion time for the building project.

(3)

(c) State one limitation in the building project used by Deva Construction Ltd.

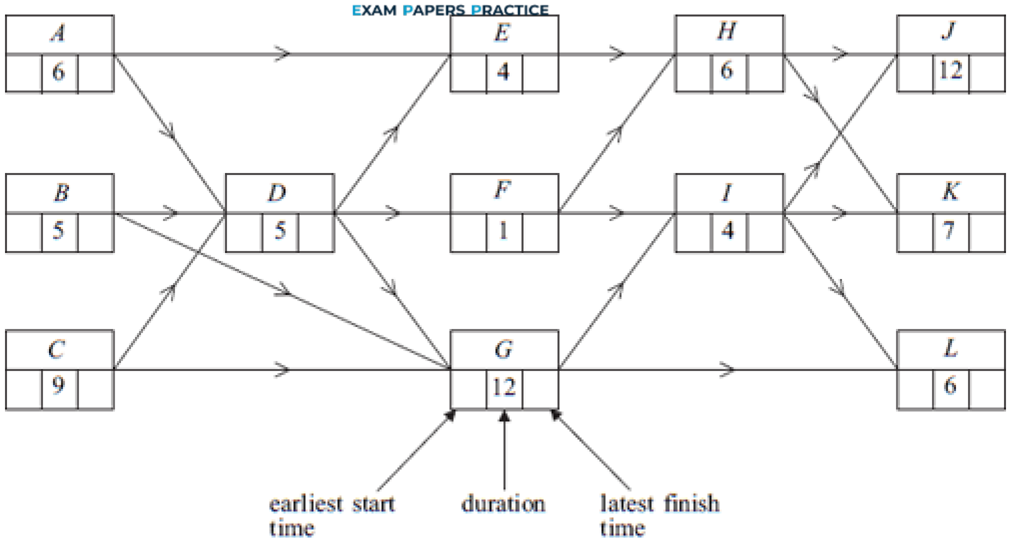
Explain how this limitation affects the project.

(2)

(Total 7 marks)

Q3.

The figure below shows an activity diagram for a project. The duration required for each activity is given in hours. The project is to be completed in the minimum time.



(a) Find the earliest start time and the latest finish time for each activity and insert their values on the figure.

(4)

(b) Find the critical path.

(1)

(c) Find the float time of activity E.

(1)

(d) Given that activities H and K will both overrun by 10 hours, find the new minimum completion time for the project.

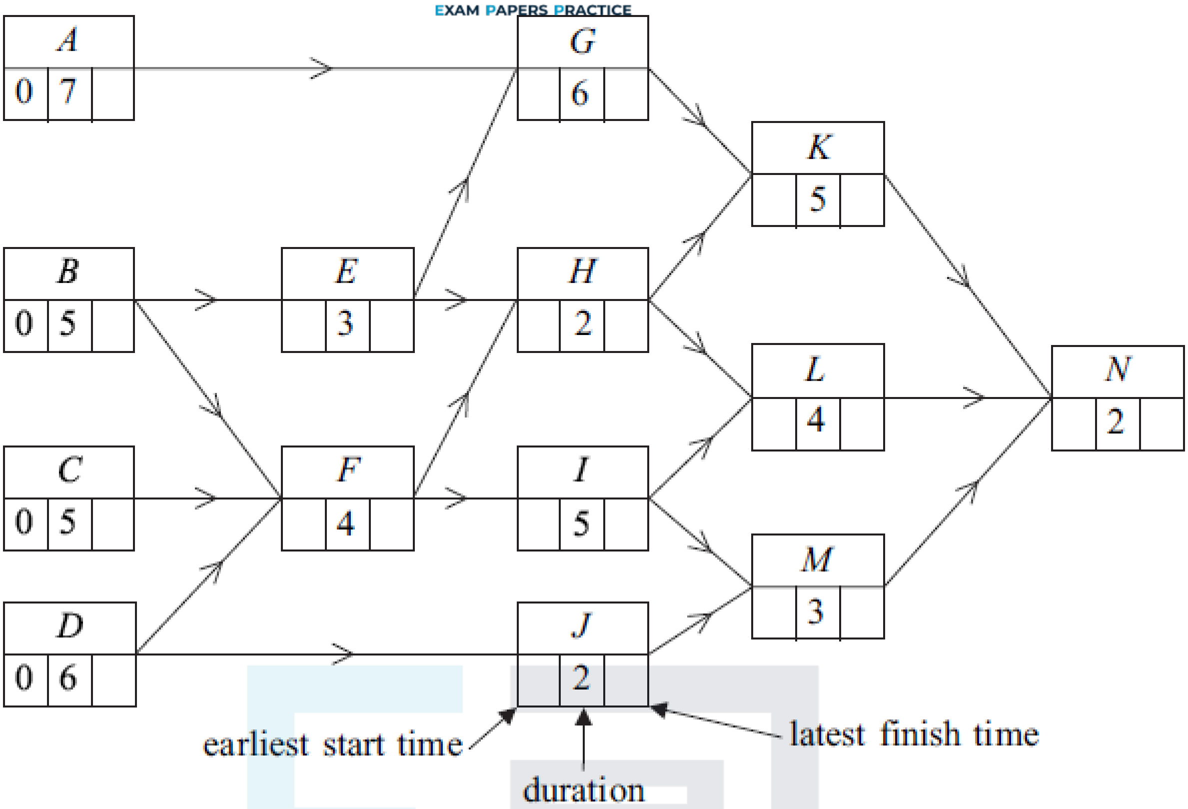
(3)

(Total 9 marks)

Q4.

Figure 1 below shows an activity diagram for a construction project. The time needed for each activity is given in days.

Figure 1



(a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

(4)

(b) Find the critical paths and state the minimum time for completion of the project.

Critical paths are ____

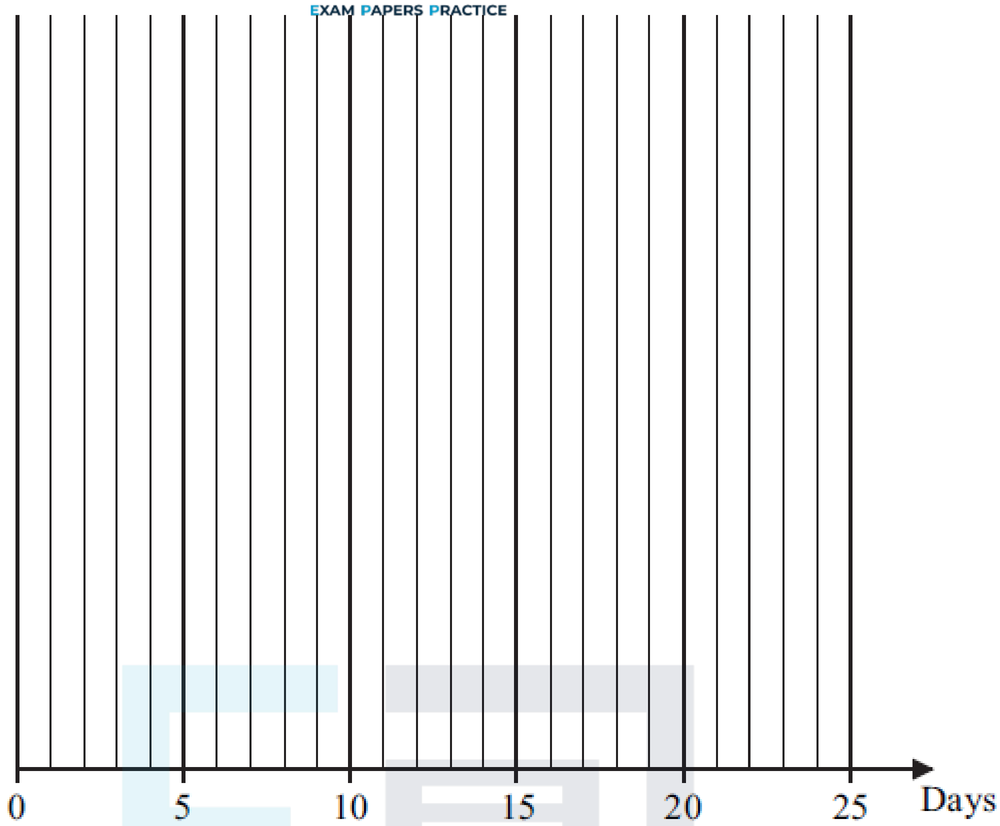
____

Minimum completion time is ______________ days.

(3)

(c) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as early as possible.

Figure 2



(5)

(d) Activity J takes longer than expected so that its duration is x days, where $x \geq 3$. Given that the minimum time for completion of the project is unchanged, find a further inequality relating to the maximum value of x.

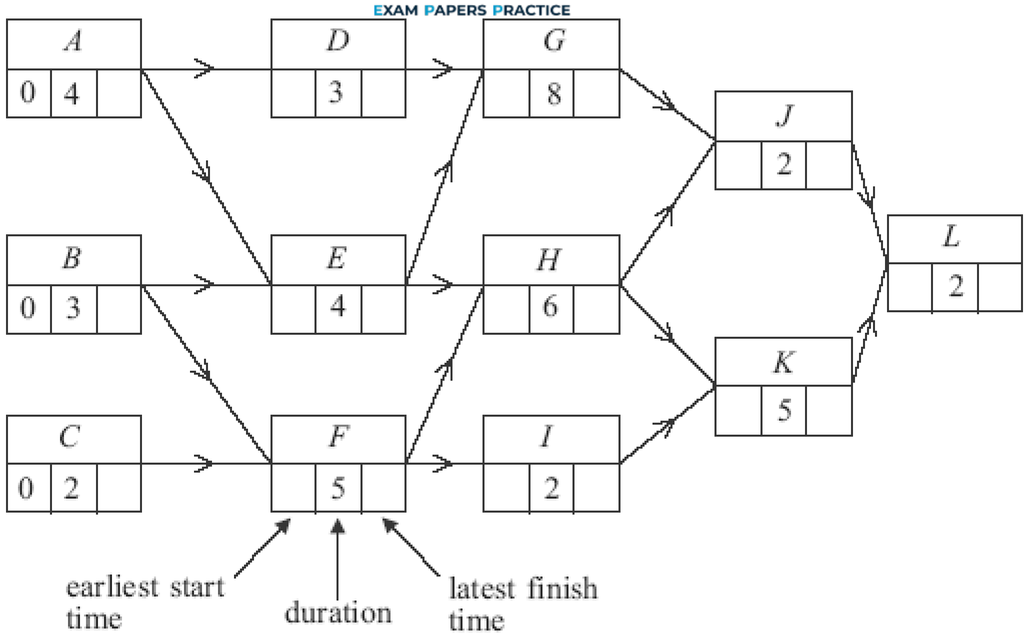
(2)

(Total 14 marks)

Q5.

Figure 1 below shows an activity diagram for a construction project. The time needed for each activity is given in days.

Figure 1



(a) Find the earliest start time and latest finish time for each activity and insert their values on **Figure 1**.

(4)

(b) Find the critical paths and state the minimum time for completion of the project.

Critical paths are ____

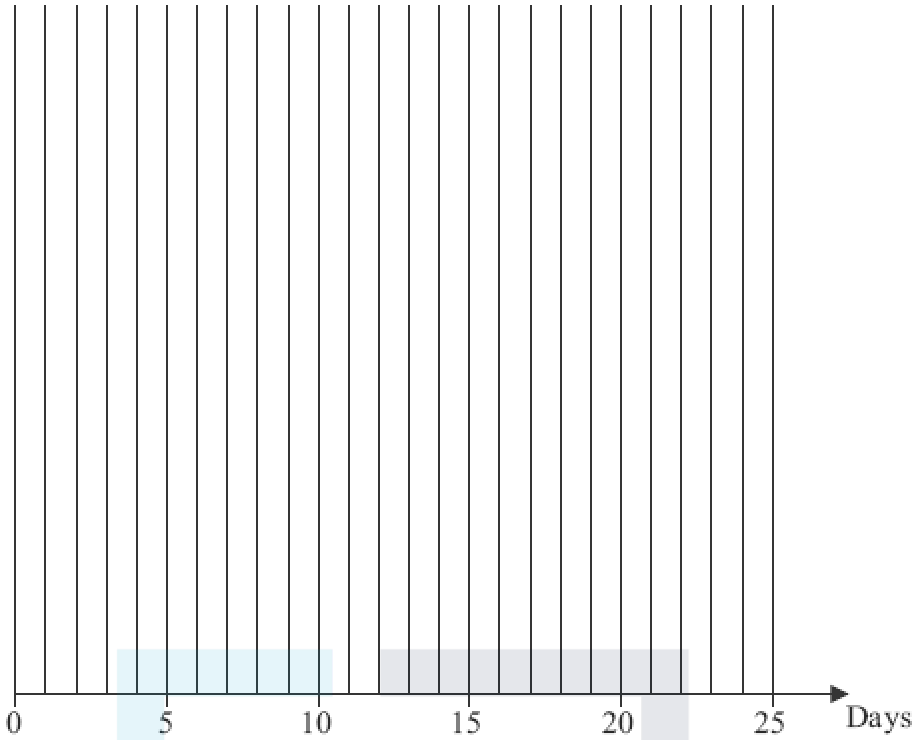
____

Minimum completion time is ____________________ days

(3)

(c) On **Figure 2** below, draw a cascade diagram (Gantt chart) for the project, assuming that each activity starts as early as possible.

Figure 2



(3)

(d) A delay in supplies means that Activity I takes 9 days instead of 2.

(i) Determine the effect on the **earliest** possible starting times for activities K and L.

(2)

(ii) State the number of days by which the completion of the project is now delayed.

(1)

(Total 13 marks)

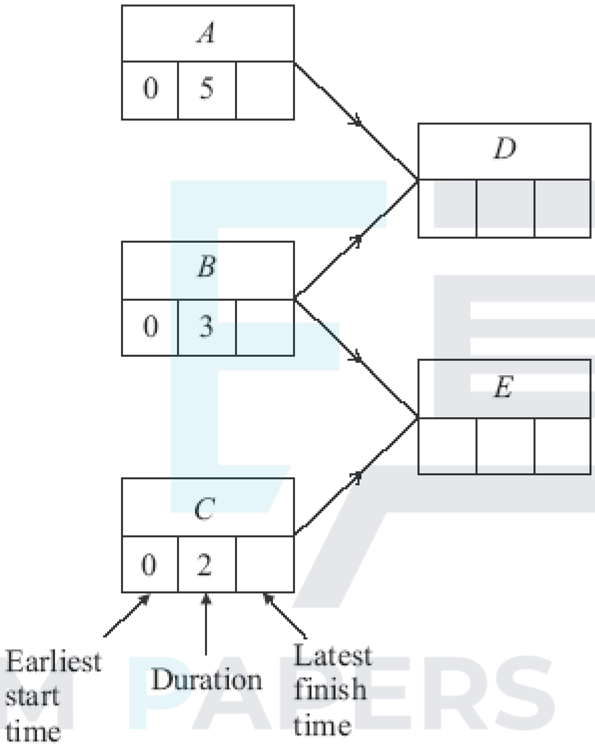
Q6.

A decorating project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (days)
A	–	5
B	–	3
C	–	2
D	A, B	4
E	B, C	1
F	D	2

G	E	9
H	F, G	1
I	H	6
J	H	5
K	I, J	2

(a) Complete an activity network for the project below.



(3)

(b) On the diagram above, indicate:

(i) the earliest start time for each activity;

(2)

(ii) the latest finish time for each activity.

(2)

(c) State the minimum completion time for the decorating project and identify the critical path.

(2)

(d) Activity F takes 4 days longer than first expected.

(i) Determine the new earliest start time for activities H and I.

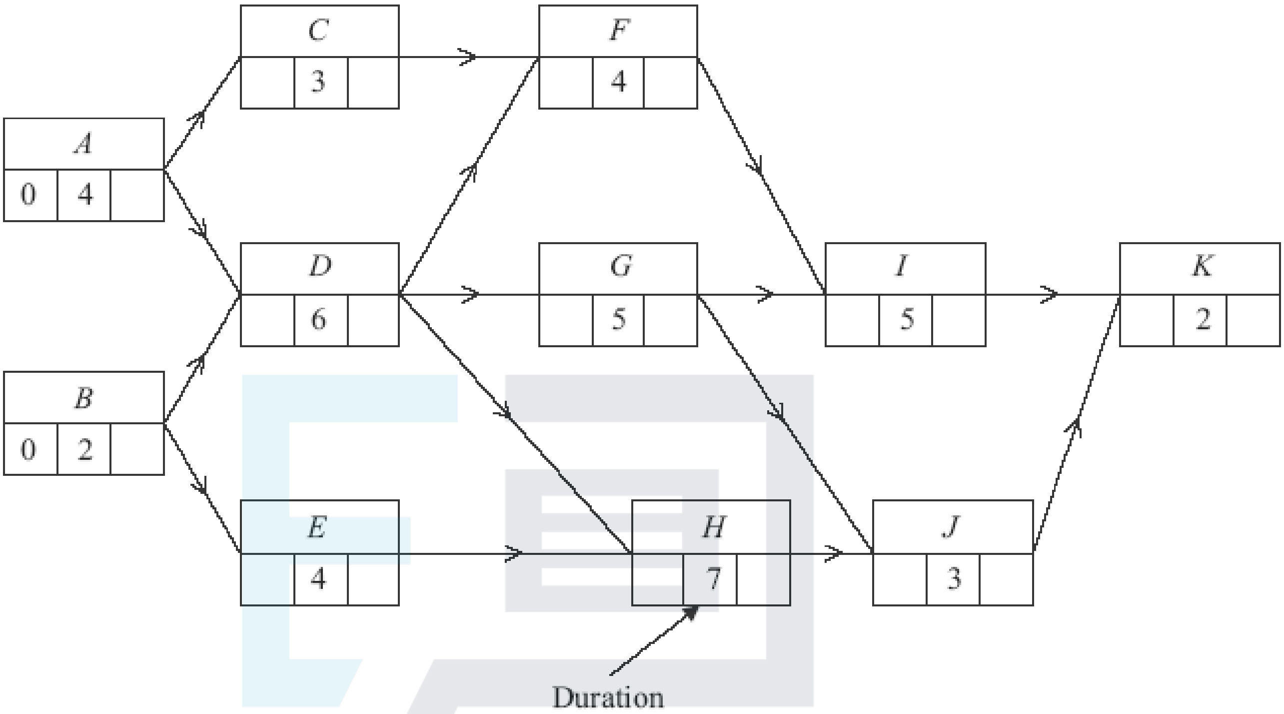
(2)

(ii) State the minimum delay in completing the project.

(1)
(Total 12 marks)

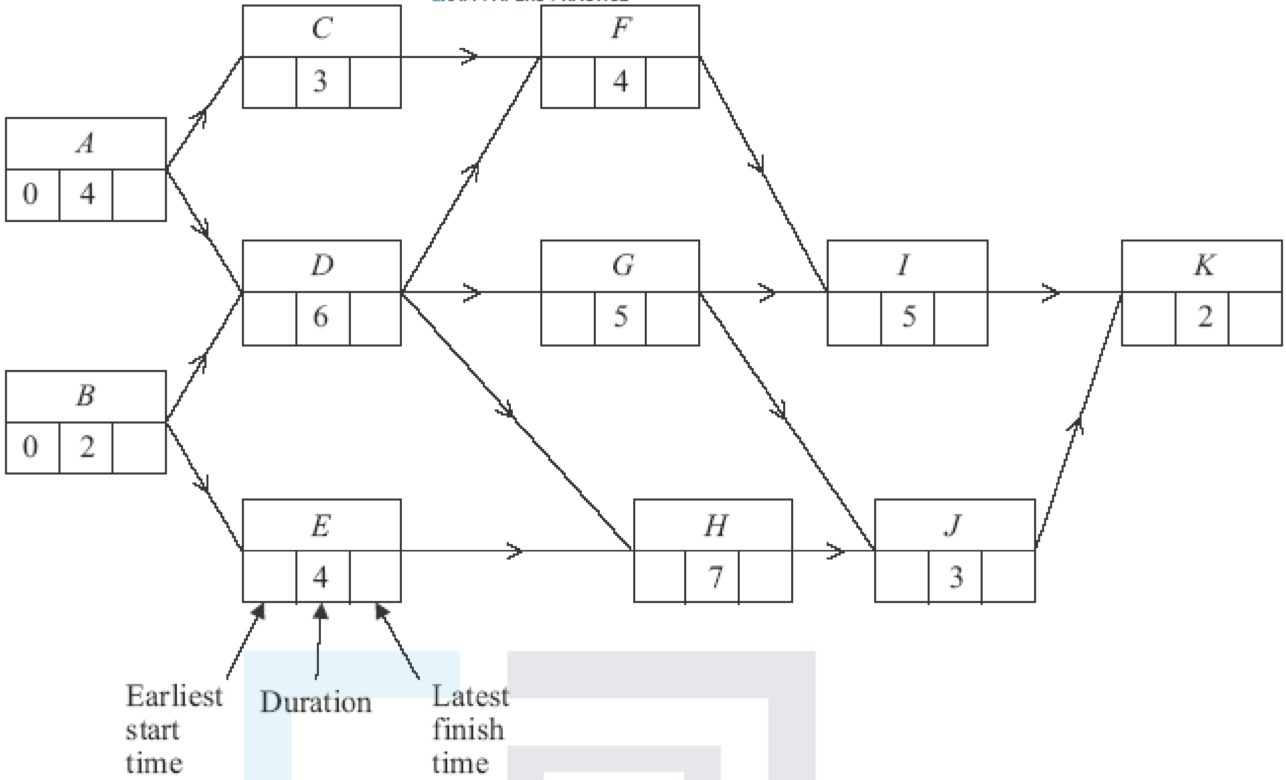
Q7.

The following diagram shows an activity network for a project. The time needed for each activity is given in days.



(a) Find the earliest start time and the latest finish time for each activity and insert their values on **Figure 1**.

Figure 1



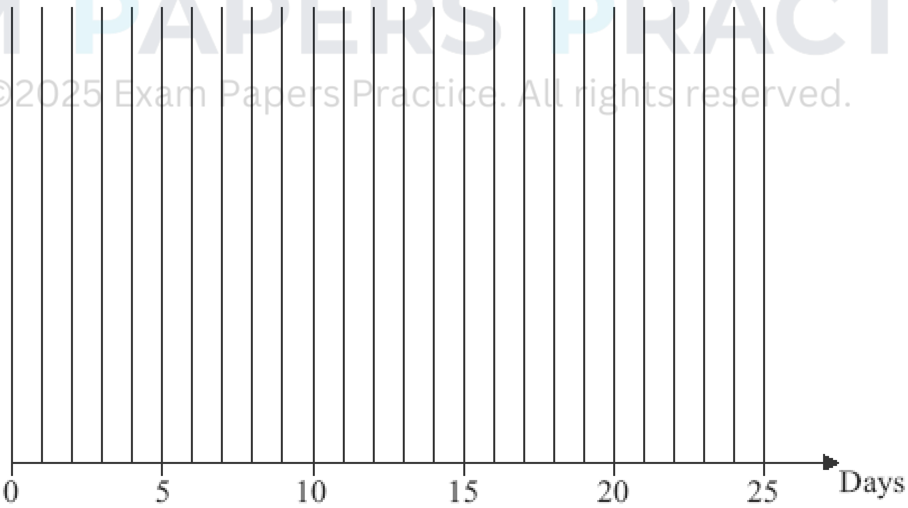
(4)

(b) Find the critical paths and state the minimum time for completion.

(3)

(c) On **Figure 2**, draw a cascade diagram (Gantt chart) for the project, assuming each activity starts as early as possible.

Figure 2



(3)

(d) Activity C takes 5 days longer than first expected. Determine the effect on the earliest start time for other activities and the minimum completion time for the project.

(2)

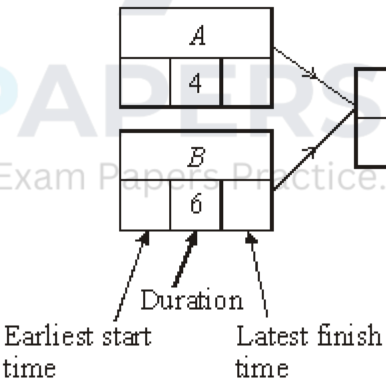
(Total 12 marks)

Q8.

A cleaning project is to be undertaken. The table shows the activities involved.

Activity	Immediate Predecessors	Duration (hours)
A	–	4
B	–	6
C	A,B	7
D	C	9
E	C	10
F	B	3
G	D,E	6
H	F,G	5
I	G	3
J	H,I	2

(a) Complete an activity network for the project on the figure below.



(4)

(b) Find the earliest start time for each activity.

(2)

(c) Find the latest finish time for each activity.

(2)

(d) (i) Write down the shortest time in which the whole cleaning project can be completed.

(1)

(ii) State which activities are the critical ones and explain how you can recognise them.

(3)

(e) A problem occurs which means that activity I cannot be started until 3 hours after activity H has started.

(i) Find the new earliest start time for activity I, giving a reason for your answer.

(2)

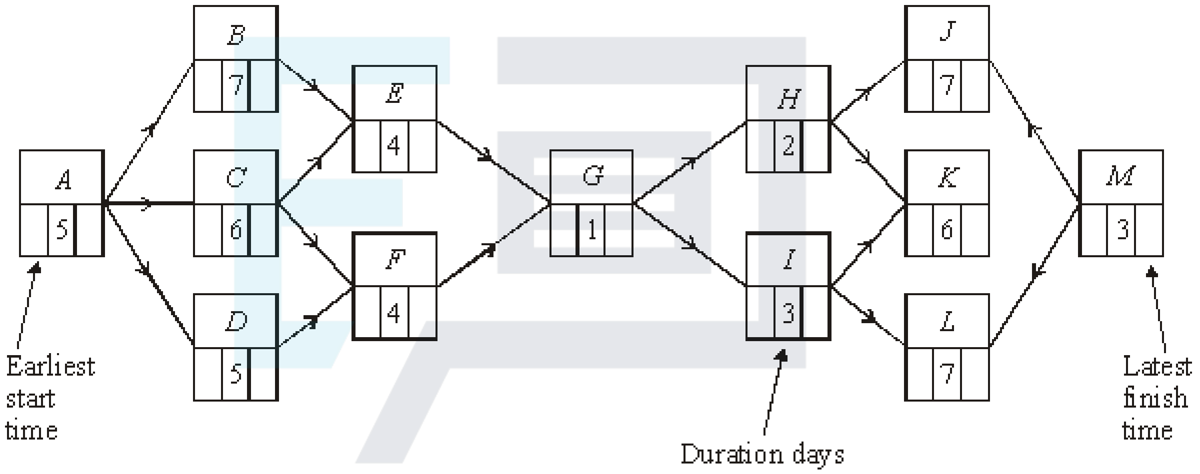
(ii) Find the revised shortest completion time for the project.

(2)

(Total 16 marks)

Q9.

The following diagram shows an activity network for a major project.



(a) On the figure above:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path.

(1)

(c) Write down the activity that has a float time of more than 1 day.

(1)

(d) Given that activity C overruns by 3 days, find the overrun time for the whole project.

(2)

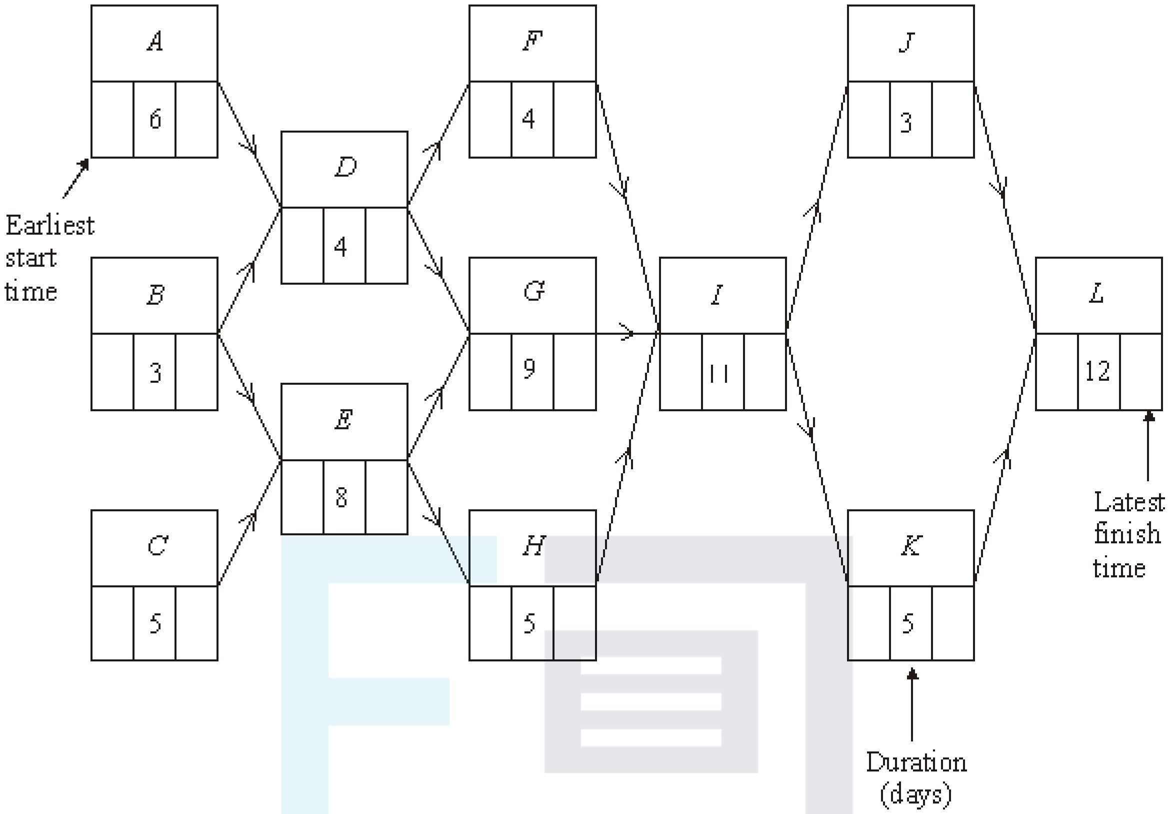
(e) Given that activities C and F overrun by 3 days each, find the overrun time for the whole project.

(3)

(Total 11 marks)

Q10.

The following diagram shows an activity network for a major building project.



(a) On the figure above find:

(i) the earliest start time for each activity;

(2)

(ii) the latest finish time for each activity.

(2)

(b) Identify the critical path.

(1)

(c) State the activity with the greatest float time.

(1)

(d) Construct a Gantt (cascade) diagram for the project.

(3)

(e) Activities D and J overrun by 5 days each.

Find:

(i) the new minimum completion time for the project;

(3)

(ii) the new critical path.

(1)

(Total 13 marks)

Q11.

A major construction project is to be undertaken and the project has been divided into 12 activities. The following diagram shows the activity network for the project.

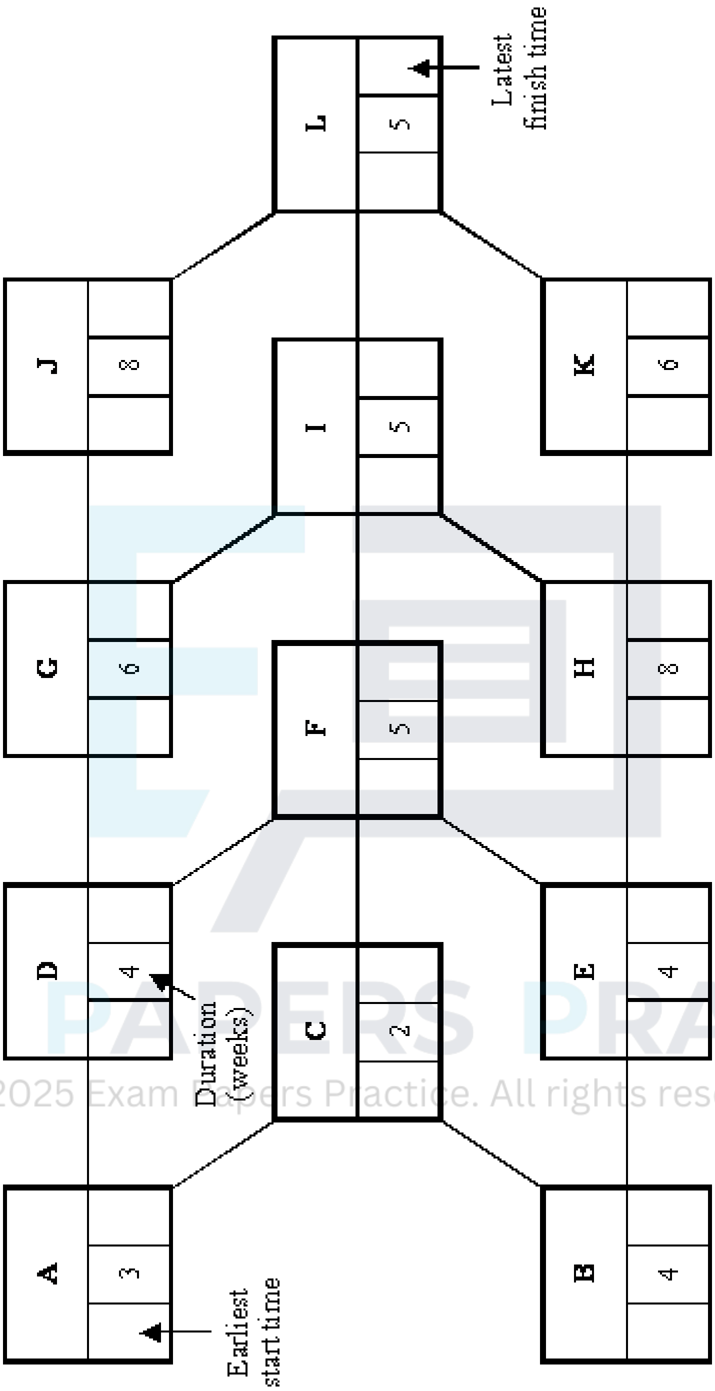


Figure 1

(a) On **Figure 1**:

(i) find the earliest start time for each activity;

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path and state the shortest completion time for the whole project.

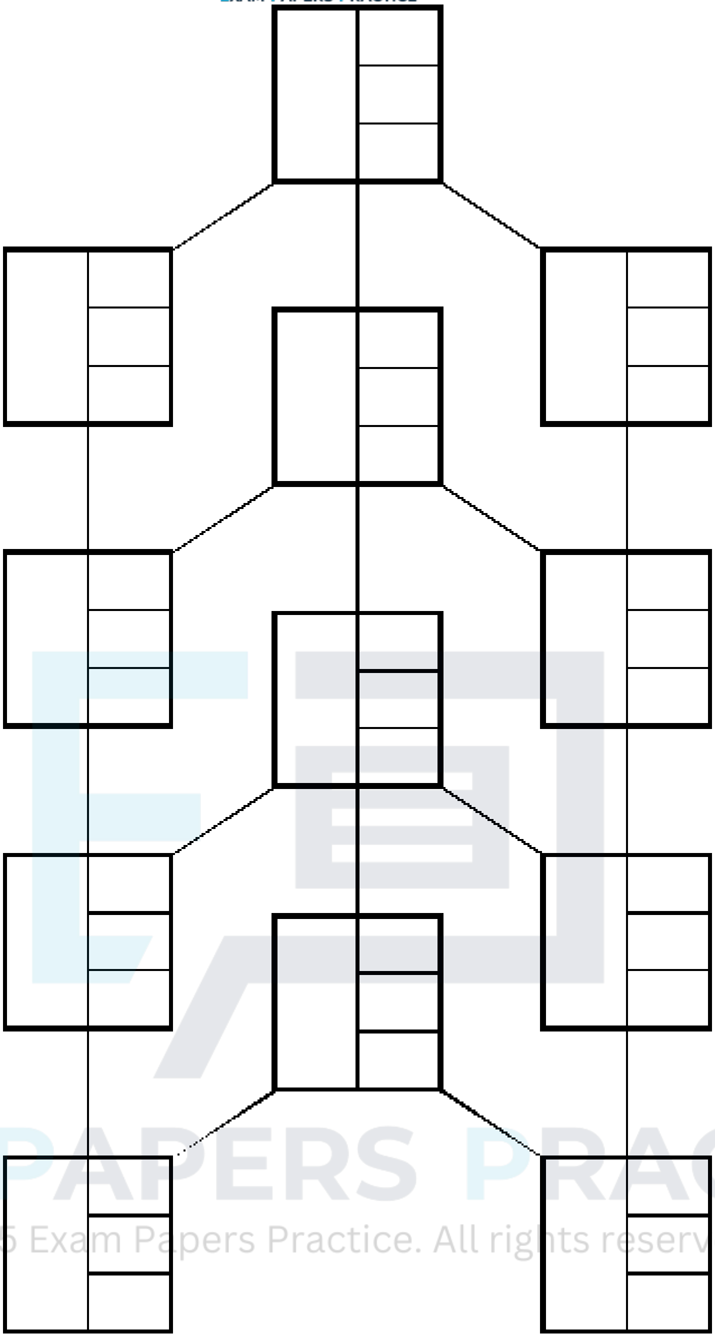
(1)

Some of the activities can be speeded up at an additional cost. The following table lists the activities that can be speeded up, together with the maximum time reduction in each case, and the extra cost for each week of reduction, in pounds per week.

Activity	**Maximum time reduction** (weeks)	**Extra cost for each week of reduction** (pounds per week)
A	1	12 000
B	2	20 000
C	1	10 000
F	1	14 000
J	3	15 000
K	4	24 000

The company wants to complete the project as soon as possible.

(c) (i) Find the revised minimum time for the completion of the whole project.
[You may use **Figure 2**, to answer this question.]



(2)

(ii) For **each** of the six activities that can be speeded up, state with justification, whether the activity should be speeded up and the reduction in the number of weeks.

(5)

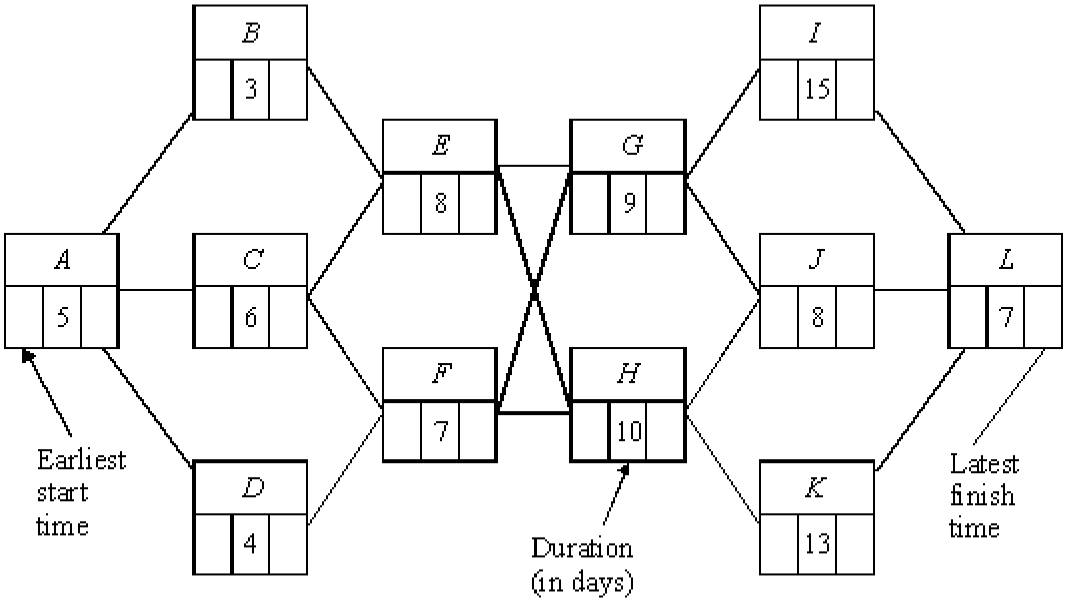
(iii) Calculate the total extra cost the company would incur in meeting this revised completion time.

(1)

(Total 13 marks)

Q12.

The following diagram shows an activity network for a major building programme.



(a) On diagram,

(i) find the earliest start time for each activity,

(2)

(ii) find the latest finish time for each activity.

(2)

(b) Identify the critical path.

(1)

(c) Construct a Gantt diagram for the project.

(3)

(d) Given that at most two activities can take place at the same time, find the new minimum completion time for the whole project

(3)

(Total 11 marks)

Q13.

The shell of a flat has been constructed and the inside is to be completed. The work has been divided into 13 activities, as shown in the table below.

Activity		Tradesman	Immediate predecessors	Duration (days)
A	Studding	Builder		2
B	Electrics (first fix)	Electrician	A	1
C	Plumbing (first fix)	Plumber	A	1.5
D	Plaster boarding	Plasterer	B, C	1

E	Plastering	Plasterer	D	1
F	Artexing	Decorator	E	1
G	Electrics (second fix)	Electrician	E	0.5
H	Joinery (first fix)	Joiner	E	1
I	Plumbing (second fix)	Plumber	E	1
J	Joinery (second fix)	Joiner	G, H, I	2
K	Painting	Decorator	F, J	1
L	Tiling	Tiler	I, J	0.5
M	Cleaning	Cleaner	K, L	0.5

(a) Construct an activity network for the project.

(4)

(b) Find the earliest start time for each activity.

(3)

(c) Find the latest finish time for each activity.

(3)

(d) Identify the critical activities and state the shortest completion time for the whole project.

(2)

(e) Construct a cascade (Gantt) diagram for the project, assuming that each activity is to start as early as possible.

(3)

(f) The foreman of the project can speed up the project by using either an extra electrician, or an extra plumber or an extra joiner.

An extra electrician would reduce the time for electrical work by 50%, an extra plumber would reduce the time for plumbing work by 25%, and an extra joiner would reduce the time for joinery work by 20%.

Explain which extra tradesman the foreman should use, and find the new minimum time to complete the project.

(5)

(Total 20 marks)

Q14.

A major construction project is to be undertaken and the project has been divided into 11 activities, as shown in the table below.

Activity	Immediate predecessors	Duration (weeks)
A	–	3
B	–	4
C	A	2
D	B	3
E	C, D	5
F	C	6
G	D	7
H	E, F, G	4
I	G	6
J	F	8
K	H, I, J	3

(a) Construct an activity network for the project.

(4)

(b) Find the earliest start time for each activity.

(3)

(c) Find the latest finish time for each activity.

(3)

(d) Identify the critical path and state the shortest completion time for the whole project.

(2)

Some of the activities can be speeded up at an additional cost. The following table lists the activities that can be speeded up, their additional cost in £ per week, together with the minimum times required to complete the activities.

Activity	Additional cost (£ **per week)**	Minimum time (weeks)
E	5000	1
I	4000	4
J	3000	5

The company wants to complete the project as soon as possible.

(e) (i) Find which activities should be speeded up. For **each** such activity, state, with justification, the reduction in the number of weeks.

(4)

(ii) Hence state the revised minimum time for the completion of the whole project.

(1)

(iii) Calculate the total additional cost the company would incur in meeting this revised completion time.

(2)

(Total 19 marks)

Q15.

A garden is to be completely renovated with the project being broken down into the activities shown in the table below.

Activity	Description	Duration (days)	Immediate Predecessors
A	Clear site	2	-
B	Prepare base for summer house	2	A
C	Prepare base for pergola	3	A
D	Erect summer house	2	B
E	Erect pergola	3	C
F	Prepare patio	2	C
G	Lay stepping stones	3	D, E
H	Build rockery	1	D, E
I	Lay patio	2	F
J	Plant out	1	H, I
K	Tidy site	2	G, J

(a) Construct an activity network for the project.

(4)

(b) Find the earliest start time for each event.

(3)

(c) Find the latest finish time for each event.

(3)

(d) Identify the critical path and state the shortest completion time for the whole project.

(2)

(e) If one of the activities overruns by two days, but the completion time is unaffected, list the activities which could overrun.

(2)

(f) If only two workers were available, each being completely responsible for the activities they undertake,

(i) state the minimum time to complete the project, giving a reason for your answer;

(2)

(ii) state how the activities could be divided between the two workers to achieve this time.

(1)

(Total 17 marks)

Q16.

The table shows the activities involved in a project, their durations, and their immediate predecessors.

Activity	A	B	C	D	E	F	G	H
Duration (days)	3	1	1	4	5	2	2	2
Immediate predecessors	–	–	B	A	A, C	B	F	D, E

(a) Draw an activity network for the project.

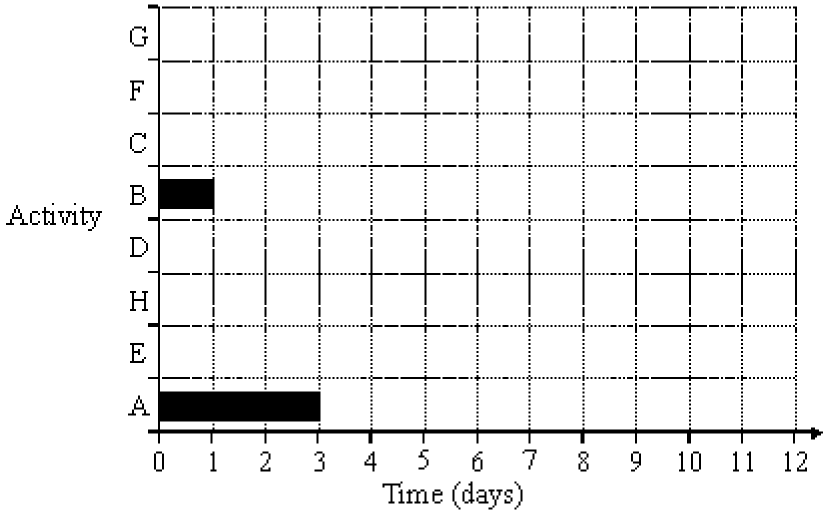
(5)

(b) Perform a forward pass and a backward pass on your activity network. Give the minimum completion time and the activities forming the critical path.

(6)

(c) In a cascade chart each activity is represented by a bar showing when it is scheduled. In the chart below, activities A and B are shown, both starting as early as possible.

Copy and complete the cascade chart, with the activities ordered as shown, and with each activity starting as early as possible.



(3)

(d) The number of people needed for each activity is as follows:

Activity	A	B	C	D	E	F	G	H
People	3	2	2	2	3	1	1	4

Activities are to be scheduled, some later than their earliest possible start time, so that at most 5 people are needed at any one time, whilst the project is still completed in the minimum time.

What are the new scheduled start times for activities D, F and G?

(2)
(Total 16 marks)